

# Appendix: Artifact Description

## Artifact Description (AD)

Paper ID: pap142

Requested SC26 badge review: all three artifact badges.

### I. CONTRIBUTIONS AND ARTIFACTS

#### A. Main Contributions

- C1: Target Dependency Packets (TDPs) expose target-centric temporal dependencies and branch-level parallelism.
- C2: DDTG constructs TDPs on demand to reduce target-irrelevant work and data movement.
- C3: OATS and PHLE reuse overlapping dependency packets while preserving ordered target updates.
- C4: TempGNN is evaluated on an Alveo U280 across six temporal datasets and four TGNN models, including accelerator-baseline comparisons.

#### B. Computational Artifacts

TABLE I  
ARTIFACT-TO-RESULT MAP.

| ID | Contents                                                                   | Contributions and results                                       |
|----|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| A1 | Four U280 sources, xclbins, hosts, inputs, tests, and measurement workflow | C1-C4; U280 latency behavior used in the accelerator comparison |
| A2 | results/result.csv and the CSV/SVG generator                               | C1-C4; Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14             |

Artifact: [10.5281/zenodo.21417187](https://zenodo.org/record/21417187); source: [TempGNN-Program/TempGNN](#); tag: sc26-ae-pap142-v1.0.

### II. DOWNLOAD

```
git clone https://github.com/TempGNN-Program/TempGNN.git
cd TempGNN
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

### III. ENVIRONMENT AND INPUTS

Board: AMD/Xilinx Alveo U280,  
→ xcu280-fsvh2892-2L-e  
Platform: xilinx\_u280\_gen3x16\_xdma\_1\_202211\_1  
OS: Ubuntu 22.04 x86\_64  
XRT: 2.16.204  
Vitis/Vivado: 2023.2 (xclbin rebuild only)  
GCC: 11.4.0  
GNU Make: 4.3  
Python: 3.10 or newer

XRT is available from <https://github.com/Xilinx/XRT>; Python is available from <https://www.python.org/>. A reviewer account on the matching U280 host is provided through the private SC26 submission channel.

A1 uses bundled prefixes of WK, MC, RT, LM, WT, and GT with JODIE, TGN, TGAT, and APAN. Input source meta-data and hashes are under external/u280\_dataset\_samples/. A2 uses results/result.csv.

## IV. RUN ON U280

A1 exercises the TempGNN mechanisms and U280 evaluation supporting C1-C4.

```
source /opt/xilinx/xrt/setup.sh
make ae-core-u280 U280_CORE_DEVICE=0
→ U280_CORE_REPETITIONS=3
```

Workflow:

- 1) check four U280 artifacts;
- 2) prepare the six-dataset/four-model inputs;
- 3) run TempGNN, MATG, ViTeGNN, and RTGA three times;
- 4) write raw measurement CSVs; and
- 5) aggregate mean latency.

TABLE II  
EXPECTED U280 MEAN LATENCY.

| Implementation | Latency (ms) |
|----------------|--------------|
| TempGNN        | 1.296553     |
| MATG           | 9.966618     |
| ViTeGNN        | 2.869752     |
| RTGA           | 4.192824     |

A successful run exits with status zero, records all four implementations, and writes fresh measurements under results/reviewer\_u280\_runs/<run-id>/. TempGNN should retain the lowest mean latency and remain close to 1.3 ms.

TABLE III  
EXPECTED A1 REVIEW TIME.

| Setup     | Execution | Analysis |
|-----------|-----------|----------|
| 10-20 min | 30-90 min | <5 min   |

Rebuilding xclbins is optional and is not part of the direct review workflow.

## V. GENERATE FIGURES FROM RESULT.CSV

A2 regenerates the paper-result plots associated with C1-C4.

```
python3 -m scripts.reproduce_paper_figures
```

The command reads results/result.csv, validates and groups the rows, and generates per-figure CSV files, matching SVG figures, and a manifest. Expected output is nine CSV/SVG pairs for Fig.2, Fig.4(a), Fig.9(b), and Fig.10-Fig.14 under results/paper\_reproduction/. The manifest must identify results/result.csv as the input.

TABLE IV  
EXPECTED A2 REVIEW TIME.

| Setup  | Execution | Analysis |
|--------|-----------|----------|
| <5 min | <1 min    | 2-5 min  |

## VI. BASELINE STATUS

- MATG: independently reproduced from the IPDPS 2022 paper and its partial public HLS headers; includes a distinct source, runner, and U280 xclbin.
- ViTeGNN: independently reproduced from the TPDS 2025 paper; includes a distinct source, runner, and U280 xclbin.
- RTGA: independently reproduced from the DAC 2024 paper; includes a distinct source, runner, and U280 xclbin.
- Cascade and TGLite-CPU: retained only as packaged paper comparison records; no fresh execution of those external stacks is claimed.
- TempGNN: includes HLS kernels, testbenches, Vitis build scripts, XRT host code, U280 board logs, and its own U280 xclbin.

The three baselines are independent, paper-based forward-path reproductions, not the authors' complete original stacks.

## VII. METRIC MEANING

DDTC is dependency-driven TDP construction. OATS is overlap-aware TDP synchronization.

`packet_reuse_factor` measures how many per-target packet materializations collapse into unique PHLE packets. `memory_bytes` is packet state/metadata traffic. `cycles` is the kernel's cycle proxy used by C-sim and exported fixture stats. HLS reports provide C/RTL latency after synthesis/cosim.

## VIII. LICENSE

This artifact is licensed under the Apache License 2.0. See LICENSE.